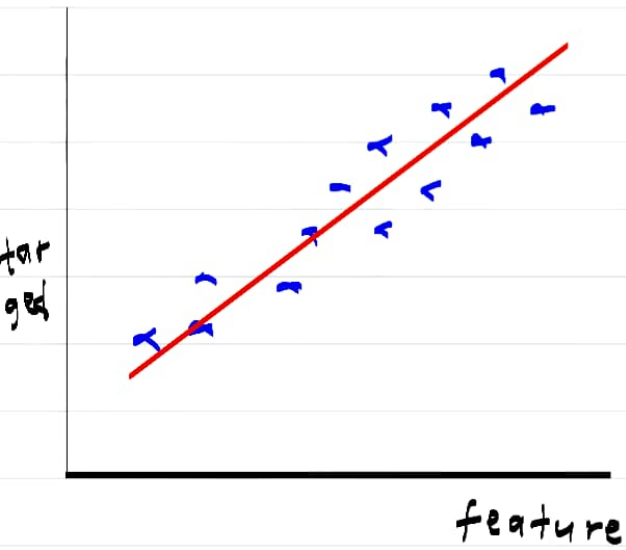
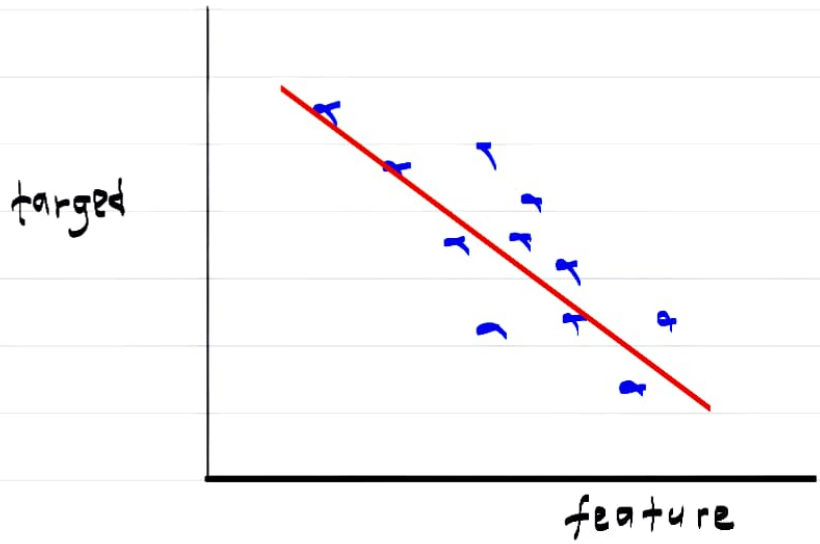
Machine Learning

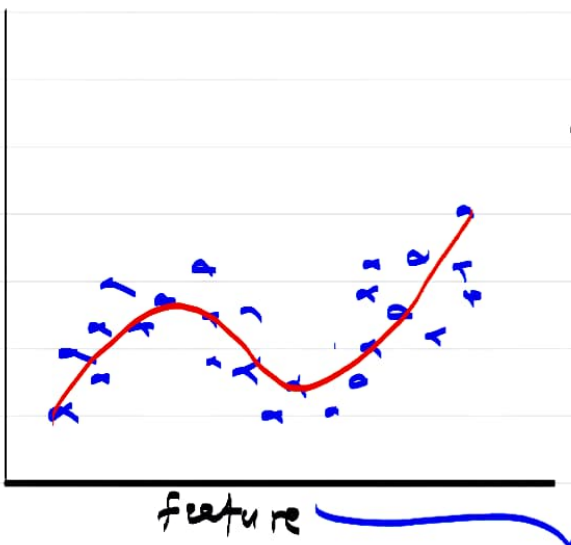
02-04-26

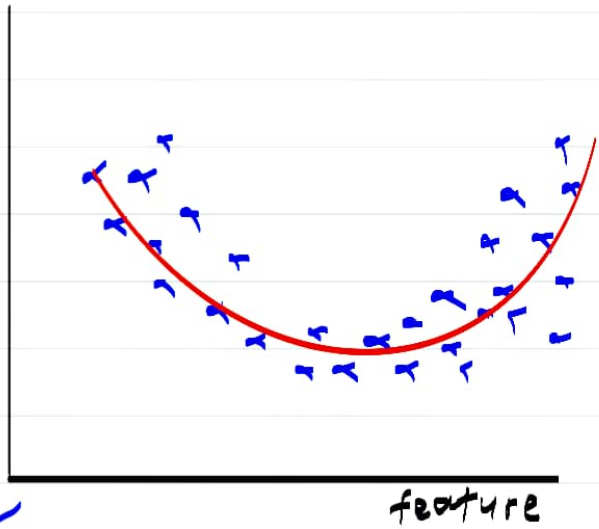
Linear Regression





in cases like above linear regression can be used



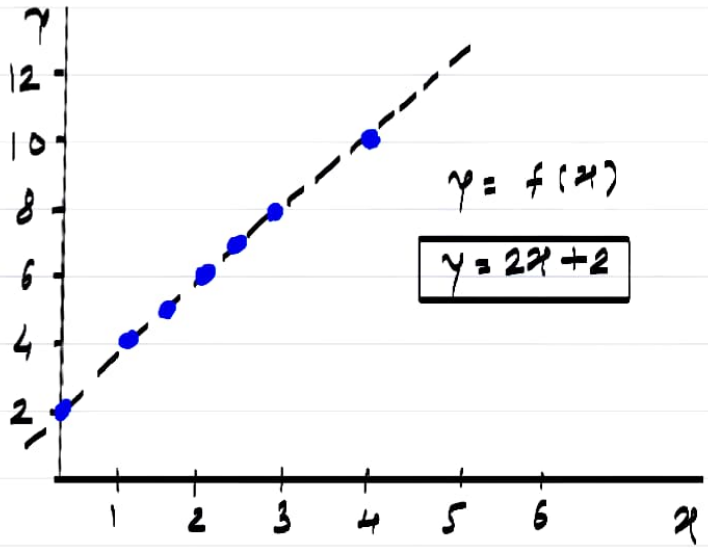


as these relationships above are not linear, linear regr will fail on these datasets

- linear relationship can be expressed using equation of straight line

eg.

x feature	y target
0	2
1	4
2	6
1.5	5
3	8
4	10
2.5	7



equation of a straight line with single feature

$$y = m x + c$$

y → output

m → slope

x → input

c → intercept / constant

this equation holds mathematical representation of relationship between input and output

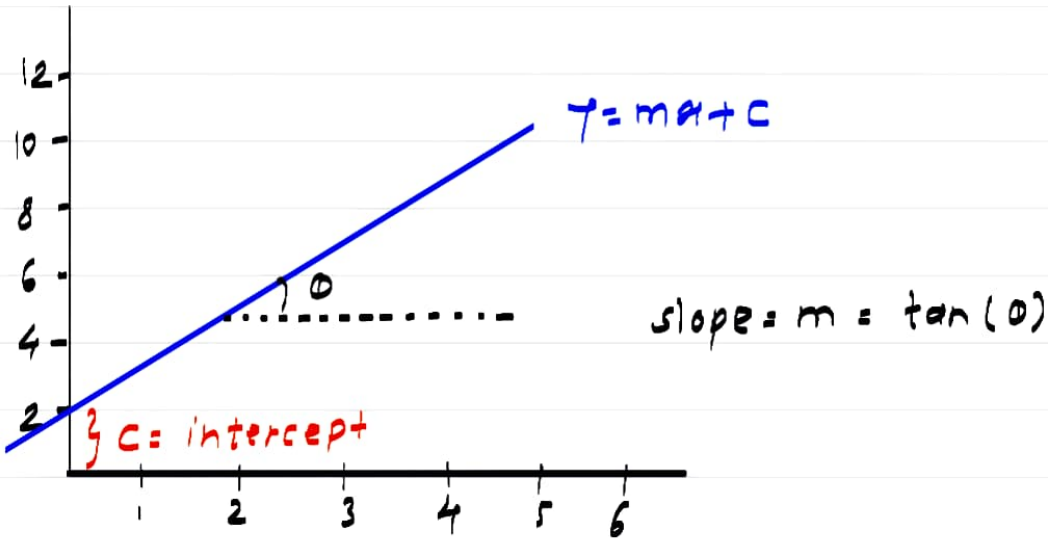
$$y = w x + b$$

y → target

w → weight

x → feature

b → bias



$$y = 2x + 2$$

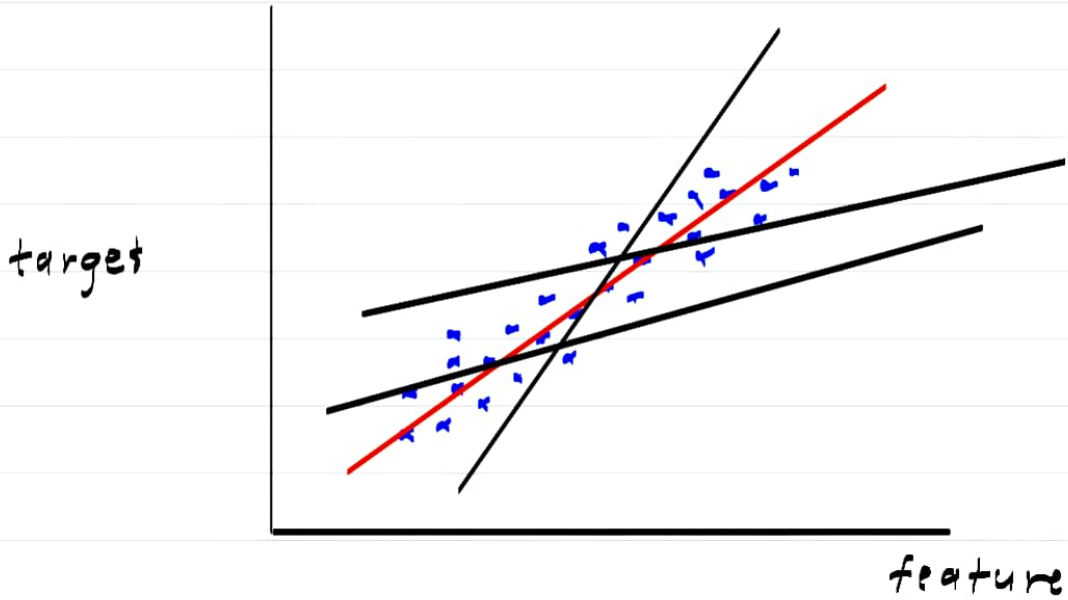
$$x = 7 \qquad y = 2 \times 7 + 2 = 16$$

Once we've found equation of this line, for any new datapoint we can find output/prediction using $y = mx + c$

$$y = wx + b \Rightarrow y = mx + c$$

- slope = m is represented as w - weight in ML
  w represents weight / contribution of feature x while computing target/prediction

- intercept - c represents b - bias in ML
  b - bias represents a value that needs to be added to equation which is independent of features



- Above dataset has approx linear relⁿ between feature and target
- This linear relⁿ can be expressed using equation of straight line
- The objective of linear regression is to find equation of straight line, which is best fit for given dataset / use case.

Best fit line - line which is best to explain given linear relⁿ

- The line with maximum accuracy
- The line with minimum error

Hypothesis function

- a function / method which converts input into output

for linear regression,

Hyp. func } $\boxed{y = wx + b}$

finding the best fit line $\Rightarrow$ finding optimal values of w and b such that we get minimum error

- the job of finding these optimal w & b values will be done by linear regr. model automatically

- w and b are called parameters of linear regr. model

- during training model tries to find optimal values for its parameters automatically

- thats why linear regr is a parametric model.

Model Parameters

- ML model tries to find optimal values of parameters automaticaly during training

- parameter values are not fixed by human

eg. w and b in linear regr. $y = wx + b$

Model Hyperparameters

- model doesn't find optimal values for these automaticaly
- Humans need to trail and error with multiple values to find optimal values for hyperparameters

eg. K - in KNN models
α - learning rate in linear regr.

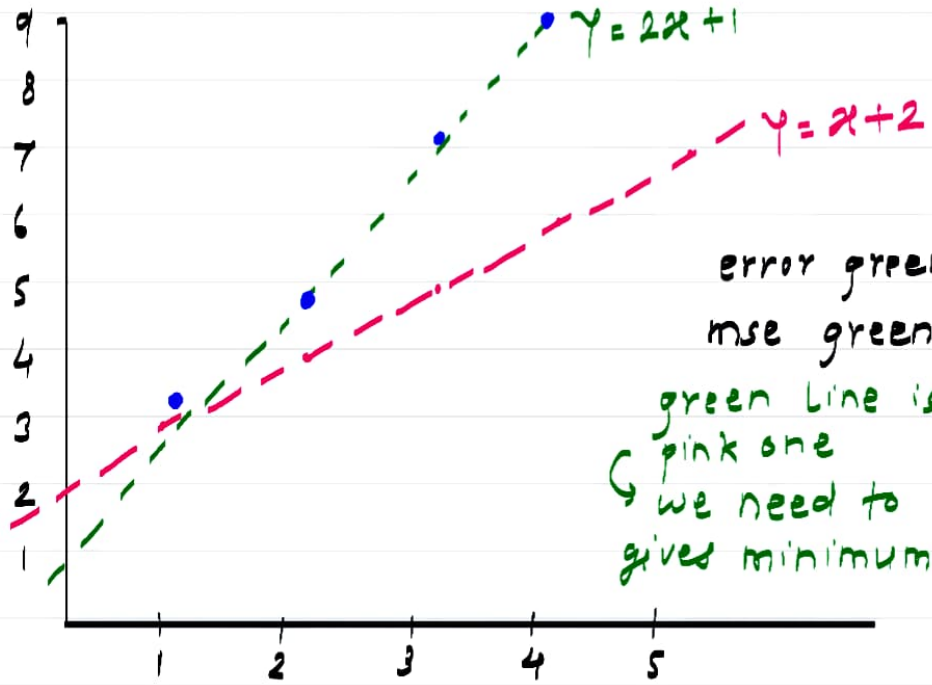
- Linear regr tries to find best fit line for given dataset

- Best fit line $\longrightarrow$ line with best/optimal values of parameters w and b

  - line with max accuracy
  - line with min error
  - line with min MSE (mean squared error)

eg.

x feature	y target
2	4.9
3	7.1
1	3.3
4	9



error green < error pink

mse green < mse pink

green line is better fit than pink one

↳ we need to find line which gives minimum mse for given data

for green line $y = 2x+1$

$x_1 = 2 \quad \hat{y}_1 = 5 \approx 4.9$

$x_2 = 3 \quad \hat{y}_2 = 7 \approx 7.1$

$x_3 = 1 \quad \hat{y}_3 = 3 \approx 3.3$

$x_4 = 4 \quad \hat{y}_4 = 9 = 9$

$y_{pred} \approx y_{act}$

for pink line $y = x+2$

$x_1 = 2 \quad \hat{y}_1 = 4 \not\approx 4.9$

$x_2 = 3 \quad \hat{y}_2 = 5 \not\approx 7.1$

$x_3 = 1 \quad \hat{y}_3 = 3 \approx 3.3$

$x_4 = 4 \quad \hat{y}_3 = 6 \not\approx 9$

$y_{pred} \not\approx y_{act}$

- find the line with minimum MSE for given data

How to do it? How to make sure mse is minimum?

☆ Loss function / error function

- MSE is widely used error / loss function for regression models

- loss function quantifies error made by model

x	y
1	3.3
2	4.9
3	7.1
4	9

step 1 - we observe approx linear rel^n and decide to train a linear regr model

step 2 - formulate hypothesis function

$$y = wx + b$$

step 3 - we start with some random values for w and b

lets say we start with $w=1$ $b=0$

this is called random initialization

any random values are fine to start with

current hyp. func $y = wx + b$

$\downarrow$

$y = x + 0 \quad (w = 1, b = 0)$

step 4- compute model predictions using current hyp. func

x	y	$\hat{y}$ $\quad y = x$
1	3.3	1
2	4.9	2
3	7.1	3
4	9	4

model predictions

step 5- compute MSE

$$mse = \frac{1}{4} \sum_{i=1}^{4} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{4} \left[(3.3-1)^2 + (4.9-2)^2 + (7.1-3)^2 + (9-4)^2 \right]$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\underline{y_{act}} - \underline{y_{pred}})^2$$

$\downarrow$ we want to minimize this

$\downarrow$ can't be changed

we can change y_{pred} to minimize mse

$\downarrow$

$y_{pred} = \underline{w}\, x + \underline{b}$

$\downarrow$ can't be changed

- w and b have to be changed to minimize mse

- change w and b values to get minimum mse.